CUSTOMER SHOPPING BEHAVIOUR ANALYSIS

---

1. Project Overview

This project examines customer shopping behaviour using transactional data consisting of 3,900 purchases across different product categories. The main objective is to generate insights into spending habits, customer segmentation, product choices, and subscription trends in order to support strategic business decision-making.

---

2. Dataset Summary

- **Rows:** 3,900
- **Columns:** 18

Key Features:

- **Customer Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Item Purchased, Category, Purchase Amount, Season, Size, Color
- **Shopping Behaviour:** Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type
- **Missing Data:** 37 missing values identified in the Review Rating column

---

3. Exploratory Data Analysis using Python

The analysis started with data preparation and cleaning using Python:

● **Data Loading:** The dataset was imported using the pandas library.

● **Initial Exploration:** Functions such as `df.info()` were used to understand the dataset structure, while `df.describe()` provided summary statistics.

● **Handling Missing Values:** Missing values in the Review Rating column were identified and filled using the median rating for each product category.

● **Column Standardization:** Column names were converted into snake_case format to improve readability and consistency.

● **Feature Engineering:**

○ An **age_group** column was created by grouping customer ages into ranges.

○ A **purchase_frequency_days** column was generated based on purchase frequency data.

● **Data Consistency Check:** It was verified whether *discount_applied* and *promo_code_used* contained duplicate information, and *promo_code_used* was removed.

● **Database Integration:** The cleaned dataset was connected to **Microsoft SQL Server**, and the processed data was loaded into the database for SQL-based analysis.

```
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   customer_id             3900 non-null   int64
 1   age                     3900 non-null   int64
 2   gender                  3900 non-null   str
 3   item_purchased          3900 non-null   str
 4   category                3900 non-null   str
 5   purchase_amount         3900 non-null   int64
 6   location                3900 non-null   str
 7   size                    3900 non-null   str
 8   color                   3900 non-null   str
 9   season                  3900 non-null   str
 10  review_rating           3900 non-null   float64
 11  subscription_status     3900 non-null   str
 12  shipping_type           3900 non-null   str
 13  discount_applied        3900 non-null   str
 14  previous_purchases      3900 non-null   int64
 15  payment_method          3900 non-null   str
 16  frequency_of_purchases  3900 non-null   str
 17  age_group               3900 non-null   category
 18  purchase_frequency_days 3900 non-null   int64
dtypes: category(1), float64(1), int64(5), str(12)
memory usage: 552.6 KB
None
        customer_id          age gender  ... frequency_of_purchases    age_group  purchase_frequency_days
count   3900.000000  3900.000000   3900  ...                    3900         3900              3900.000000
unique          NaN          NaN      2  ...                       7            4                      NaN
top             NaN          NaN   Male  ...          Every 3 Months  Young Adult                      NaN
freq            NaN          NaN   2652  ...                     584         1028                      NaN
mean    1950.500000    44.068462    NaN  ...                     NaN          NaN                89.133077
std     1125.977353    15.207589    NaN  ...                     NaN          NaN               119.037566
min        1.000000    18.000000    NaN  ...                     NaN          NaN                 7.000000
25%      975.750000    31.000000    NaN  ...                     NaN          NaN                14.000000
```

4. Data Analysis using SQL (Business Transactions)

Structured analysis was carried out in **Microsoft SQL Server** to address important business questions:

1. **Revenue by Gender** – Compared the total revenue contributed by male and female customers.

	GENDER	TOTAL_REVENUE
1	Male	157890
2	Female	75191

2. **High-Spending Discount Users** – Identified customers who applied discounts but still spent more than the average purchase amount.

	CUSTOMER_ID	PURCHASE_AMOUNT
87	176	86
88	177	81
89	181	81
90	182	61
91	183	96
92	188	94
93	190	82
94	191	70
95	192	76
96	194	100
97	195	94

3. **Top 5 Products by Rating** – Determined the products with the highest average customer ratings

	ITEM_PURCHASED	AVG_RATING
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.8
5	Handbag	3.78

4. **Shipping Type Comparison** – Analyzed the difference in average purchase values between Standard and Express shipping methods.

	SHIPPING_TYPE	AVG_AMOUNT
1	Standard	58
2	Express	60

5. **Subscribers vs. Non-Subscribers** – Compared both average spending and total revenue based on subscription status.

	SUBSCRIPTION_STATUS	TOTAL_CUSTOMERS	AVG_SPENT	TOTAL_REVENUE
1	No	2847	59	170436
2	Yes	1053	59	62645

6. **Discount-Dependent Products** – Identified five products with the highest proportion of discounted purchases.

	ITEM_PURCHASED	DISCOUNT_RATE
1	Hat	50
2	Coat	49
3	Sneakers	49
4	Sweater	48
5	Pants	47

7. **Customer Segmentation** – Categorized customers into New, Returning, and Loyal segments based on their purchase history.

1	RETURNING	701
2	LOYAL	3116
3	NEW	83

8. **Top 3 Products per Category** – Listed the most frequently purchased products within each category.

	ITEM_RANK	CATEGORY	ITEM_PURCHASED	TOTAL_ORDERS
1	1	Accessories	Jewelry	171
2	2	Accessories	Belt	161
3	3	Accessories	Sunglasses	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Evaluated whether customers with more than five purchases are more likely to subscribe.

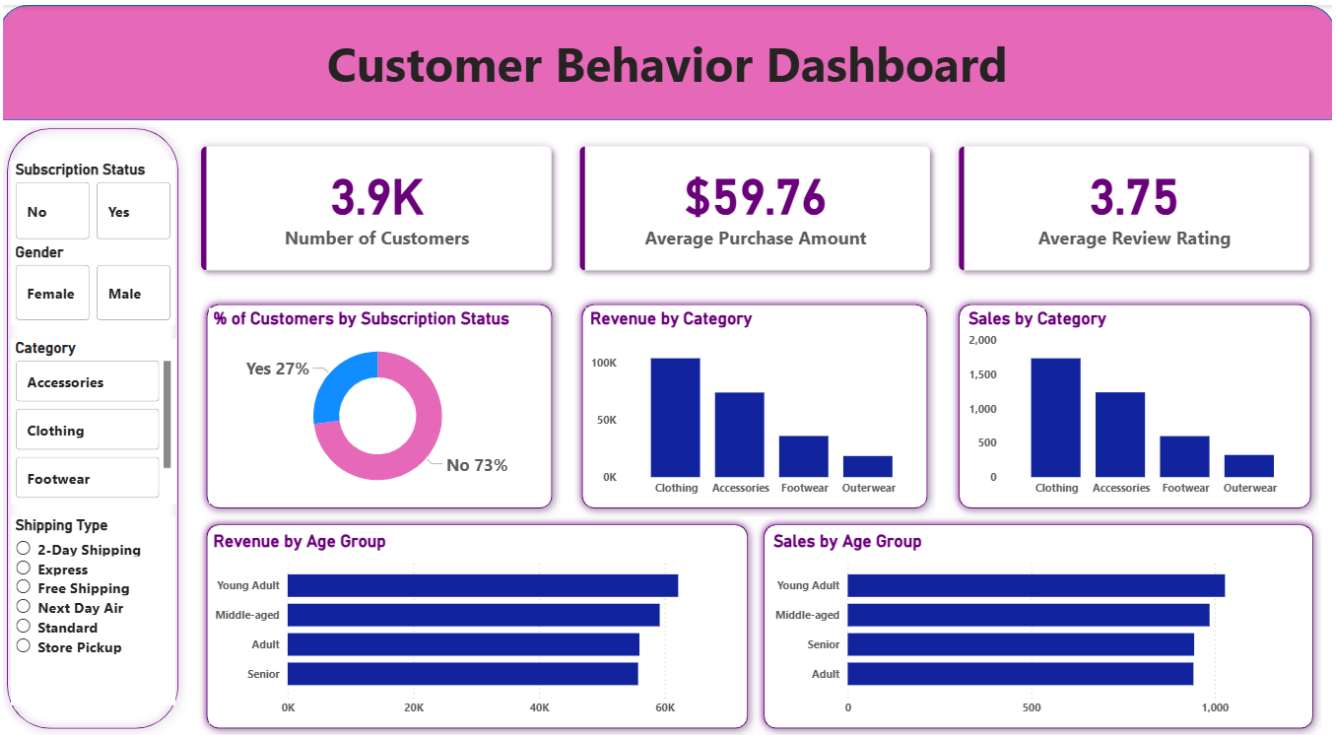
	SUBSCRIPTION_STATUS	REPEAT_BUYERS
1	Yes	958
2	No	2518

10. **Revenue by Age Group** – Calculated the total revenue generated by different age groups.

	AGE_GROUP	TOTAL_REVENUE
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

An interactive dashboard was developed in Power BI to visually represent the insights obtained from the analysis. This dashboard helps in better understanding trends, comparisons, and key performance indicators.



---

6. Business Recommendations

● **Boost Subscriptions** – Encourage subscriptions by offering exclusive benefits and incentives.

● **Customer Loyalty Programs** – Introduce reward systems to convert repeat customers into loyal customers.

● **Review Discount Policy** – Maintain a balance between increasing sales and protecting profit margins.

● **Product Positioning** – Promote top-rated and best-selling products effectively in marketing campaigns.

● **Targeted Marketing** – Focus marketing strategies on high-revenue age groups and customers who prefer express shipping.